

Common trainee mistakes — five RAG misconceptions

The five misconceptions a trainee is most likely to hold, and how to correct each — with why trainees believe it, the correction, a short example, and a teaching tip. Use these during the session (they map to Slides 3-5, 12, 15, 18-19) and as a discussion prompt in the wrap-up.

Misconception 1 — “RAG means the LLM *knows* my documents”

Why trainees believe it. After a RAG demo answers questions about their files, it feels as if the model “learned” or “absorbed” the documents — the same way they’d describe a colleague who read the wiki. The vocabulary of “training” and “knowledge” blurs the line between learning and looking something up.

Correction. Nothing about the model’s weights changes. At question time the pipeline *retrieves* relevant passages and *pastes them into the prompt* as context; the model reads that context and answers. Remove the retrieval step, or ask a question whose documents weren’t retrieved, and the “knowledge” vanishes. RAG changes **what’s in the prompt**, not **what’s in the model**.

Short example. In the lab, `lumen_cli.md` describes a **fictional** tool. The model could not have memorised it. Yet the system answers “Version 2.3 introduced the `--fast` flag” correctly — because that sentence was retrieved and placed in the context, not because the model knows Lumen CLI.

Teaching tip. Run the same question twice: once through `generate_answer()` (with retrieval) and once by asking the bare model (or just show the empty context). The contrast — correct-with-context vs. can’t-answer-without — makes “retrieved, not learned” unforgettable. Contrast RAG with fine-tuning explicitly: fine-tuning changes weights; RAG changes the prompt.

Misconception 2 — “More chunks always mean better retrieval”

Why trainees believe it. More evidence sounds safer — surely giving the model everything can’t hurt? It mirrors a human intuition (“read more before answering”) and the belief that the LLM will simply ignore anything irrelevant.

Correction. Each extra chunk you retrieve (larger `top_k`) or each oversized chunk adds **noise** as well as signal. Irrelevant or loosely-related passages dilute the context, can crowd out the one chunk that mattered, cost tokens and latency, and give the model more opportunities to anchor on the wrong passage. There’s a sweet spot; past it, quality *drops*.

Short example. The Top-K stretch challenge: for the Lumen question, `top_k=1` returns only the correct `lumen_cli.md` passage; `top_k=5` drags in unrelated chunks from `version_control.md` and `python_functions.md`. The signal is unchanged; the noise grew.

Teaching tip. Have trainees run `top_k = 1, 3, 5` and read the sources list each time. Ask: “Did a higher K ever add a *useful* chunk, or just noise?” Connect it to chunk size — a huge chunk is just a sneaky way of retrieving lots of (mostly irrelevant) text at once.

Misconception 3 — “Vector search always retrieves the correct answer”

Why trainees believe it. Semantic search feels like magic the first time “car” matches “automobile.” It’s easy to over-trust and assume the top result is always the *right* result.

Correction. Retrieval is a similarity heuristic, and it can fail at every stage: bad **chunking** (the fact was split or buried), weak **embeddings** (the model doesn’t capture the needed nuance), **query wording** that doesn’t resemble the source, **missing information** (it’s simply not in the corpus), or **ranking** that puts a near-miss on top. “Similar” is not “correct.”

Short example. Ask the lab “What is the capital of Mars?” Vector search still returns its Top-K chunks — it *always* returns something — but every score is ~0. Similarity ranked them; it did not find an answer, because there is none.

Teaching tip. Stress that retrieval **always returns K results**, even for nonsense — returning something is not the same as finding the answer. This is the bridge to the grounding guard (Misconception 4 / Slide 19): because scores are meaningful, a threshold can separate “found it” from “nearest noise.”

Misconception 4 — “RAG completely eliminates hallucinations”

Why trainees believe it. RAG is marketed as *the* fix for hallucination, and it genuinely helps, so people round “reduces” up to “eliminates.”

Correction. RAG reduces *unsupported* answers by grounding generation in retrieved text, but both halves can still fail. **Retrieval** can miss or surface the wrong passage; **generation** can over-reach, blend in parametric knowledge, mis-summarise, or ignore a qualifier. If the corpus itself is wrong, a perfectly grounded answer is confidently wrong. Grounding lowers the *rate* and makes errors *traceable* — it doesn’t make them impossible.

Short example. Give the model a correct passage but ask a question it only *partly* answers (e.g. a dose for a different population). An abstractive LLM may “helpfully” fill the gap. The repo’s production prompt guards against exactly this by quoting whole chunks and refusing partial support.

Teaching tip. Reframe the goal from “zero hallucinations” to “**grounded, attributable, and refusable**.” Ask the room: “If the retrieved document is itself wrong, what does a perfect RAG system do?” (Answer: confidently repeats the error — which is why source quality and evaluation matter.)

Misconception 5 — “If a document was retrieved, the answer must be trustworthy”

Why trainees believe it. Seeing a citation triggers the same trust we give a cited academic paper. A source *appears*, so the answer feels verified.

Correction. Retrieval is not verification. A citation proves a passage was *retrieved and used*; it says nothing about whether the passage is correct, current, or actually answers the question. The chunk was selected for **similarity**, not **truth**. Verification — is the source authoritative? up to date? on point? — is a separate step a human or a downstream check must perform.

Short example. The two synthetic review-interval memos in the production corpus state different intervals for the same task. Retrieval can surface either; a citation to one doesn’t make it the right one. The production system treats this as *conflicting evidence* and refuses rather than picking a “winner.”

Teaching tip. Teach the mantra “**provenance, not proof**.” A citation tells you *where to look*, not *that it’s right*. Have trainees articulate the difference between “the system showed its source” and “the source is correct” — then ask who, in their own product, is responsible for closing that gap.

One-line summary (for the recap slide)

#	Misconception	The correction in five words
1	The LLM “knows” my docs	Retrieved into context, not learned
2	More chunks = better	Noise grows with the signal
3	Search always finds it	Similarity can fail everywhere
4	RAG ends hallucination	It reduces, never eliminates

#	Misconception	The correction in five words
5	Retrieved = trustworthy	Provenance is not proof